Tsukuyomi Watch

A Japanese-inspired guide from observing plans to astronomical image inspection.

Choose your place.
Find your moment.
Explore the faint light.



Nighttime weather • cosmic events • local AI restoration

LIVE tsukuyomi-watch.vercel.app

A sky full of possibilities. A fragmented workflow.

Students and hobby observers must combine several tools before they can begin.

WHERE TO GO

Conditions change

Cloud cover, humidity and visibility vary across places and observing hours.

WHEN TO LOOK

Events are scattered

Moon phases, eclipses, meteor windows, launches and ISS predictions need different sources.

WHAT TO TRUST

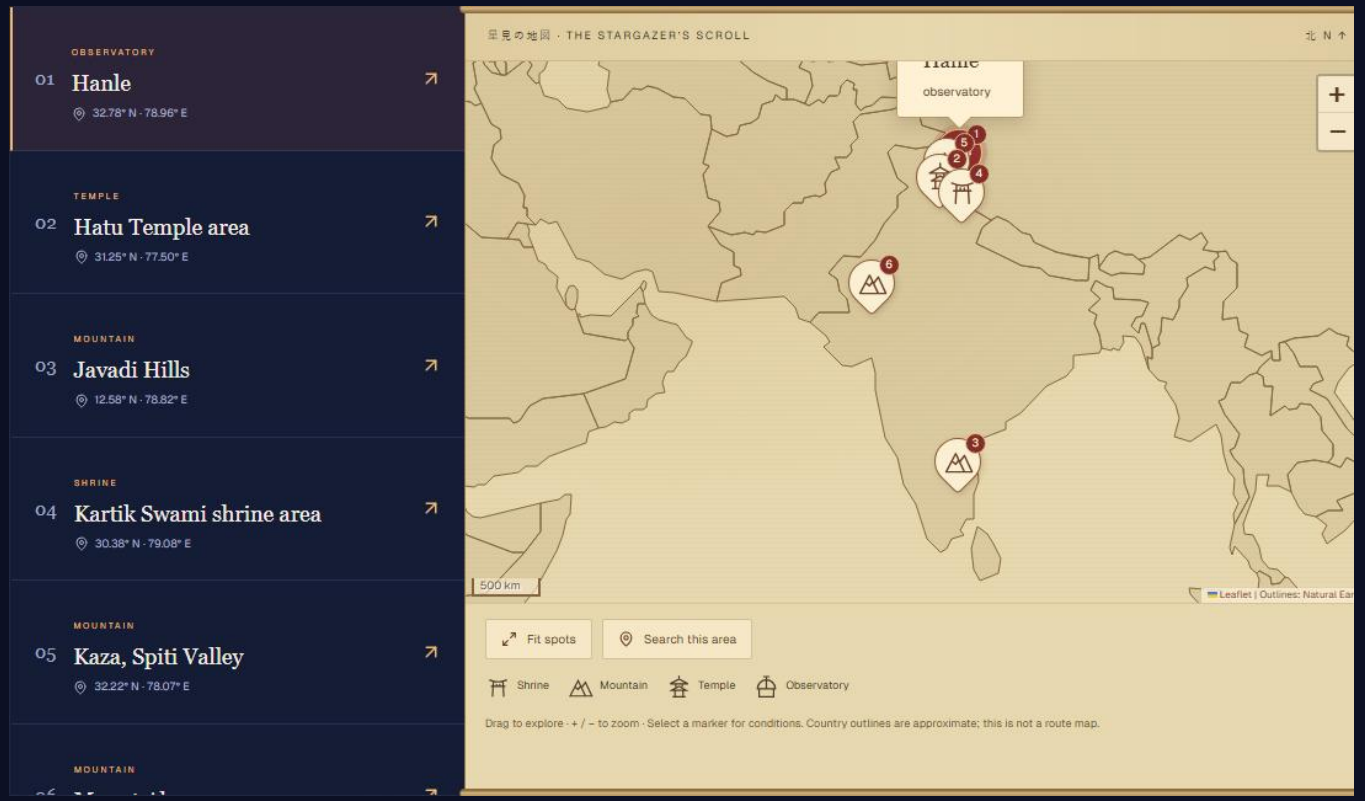
Images need inspection

Noise and blur obscure faint sources. Cleanup needs conservative edits and a clear comparison.

Designed for astronomy clubs, curious students and hobby astrophotographers.

The Watchtower + The Celestial Scroll

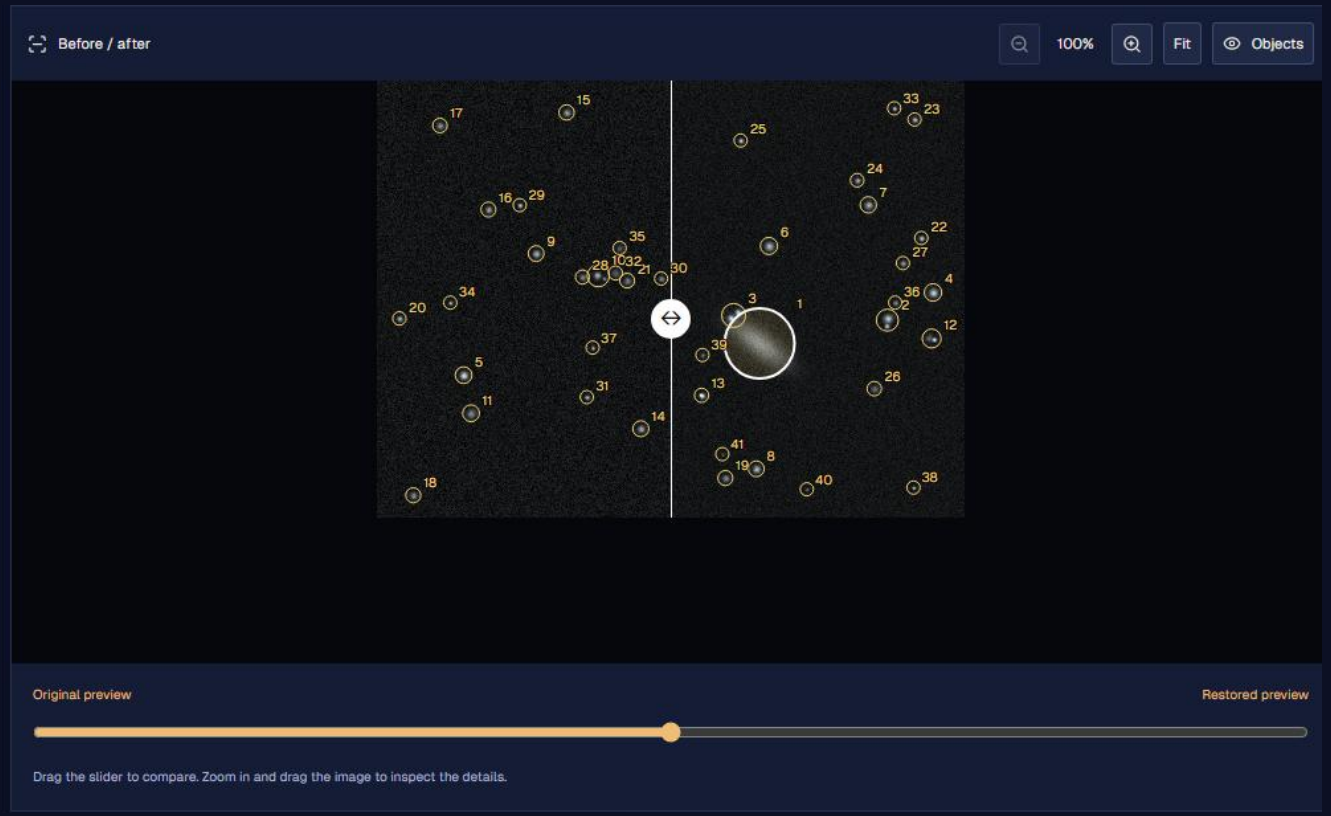
Choose a place and a moment through a scroll-style map and lunar calendar.



- Search worldwide cities; explore nearby curated observing spots.
- Compare nighttime cloud cover, humidity, visibility and a weather score.
- Browse lunar phases, launches, eclipses and meteor windows.
- Open event details, view ISS predictions and prepare reminders.

Spirit Lens + Dojo visualization

Local AI cleanup, a before/after slider, zoom and object-candidate overlays.



- Pretrained DnCNN runs in a browser worker.
- Gentle bounded edits; optional conventional deblur.
- Inspect source size, brightness and tentative morphology.
- Download the restored preview and analysis report.

A typed backend. A private local image pipeline.

Next.js / React • Express / tRPC • Zod • Drizzle / PostgreSQL • ONNX WASM

Web interface

Map, calendar, feature pages
Typed requests + React Query

→

API + domain services

Validated requests, provider adapters, caches and alert jobs

→

PostgreSQL / Neon

Destinations, forecasts, events, feeds and push jobs

Local image worker

Validate → DnCNN → bounded blend → inspect

Independent data sources

Open-Meteo, USNO, CelesTrak, NASA, launches

An observing journey with its own identity

The innovation is the connected experience and inspectable processing.

MORE WAYS TO EXPLORE

Archive + signals

Import NASA archival survey images. Screen light-curve and spectral CSVs for follow-up candidates.

PRACTICE AND RETURN

Honor journal

Log observations, earn 25 points per event and unlock badges. Keep an exportable device-local record.

JAPANESE-INSPIRED

Four visual motifs

Ink Wash, Torii, Sakura and Katana. Light/dark modes, moon cues, translated labels and navigation notes.

Established scientific methods, combined with transparent sources and a distinctive interface.

Built, deployed and tested against edge cases

A live prototype with repeatable checks and a practical club-pilot path.

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Automated cases passing	Test files	Production builds passing

- Boundary cases: missing forecasts, lunar leap months, invalid orbital data and unsafe push endpoints.
- Image safeguards, malformed CSVs, duplicate observations and global city-query behavior are tested.
- Live on Vercel + Render + Neon; local inference avoids a separate GPU inference server.

Next validation: broader E2E, physical push receipt, astronomy benchmarks and real user outcomes.

Make the first observing night easier to plan.

A connected guide for curiosity, observation and careful image exploration.

Start with a club.
Validate with real nights.
Grow the observing map.

NEXT MILESTONES

- More curated destinations
- Validated source preservation
- Reliable scheduled alerts
- Authenticated cross-device sync

Try the working demo

tsukuyomi-watch.vercel.app

github.com/PECathon-2026/Team-PF-K96JPZ